Sahaj Bhadja

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Summary

Software Engineer with 2+ years building and validating **connected software systems**, **testable backend workflows**, and **cloud-integrated applications** using Python, C++/Java, SQL, CI/CD, and cross-functional Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Developed and validated **connected data workflows** for 500+ users on **Azure SQL/PostgreSQL**, reducing latency from **1.2s to 350ms** through optimization and reliability-focused debugging.
- Implemented **release validation pipelines** for 5K+ monthly jobs with reconciliation checks, improving responsiveness, fault detection, and production readiness for user-facing software components.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **Python simulation/test pipelines** processing 100K+ trajectory logs, enabling reproducible validation and performance tuning of complex systems in iterative engineering cycles.
- Produced structured defect analysis and test documentation, accelerating issue triage and improving cross-team collaboration across software, systems, and model-validation stakeholders.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL integrations** with OOP, AWS Glue, and S3, improving response times by **30%** and supporting stable cloud-connected service behavior.
- Maintained **20+ unit/integration test suites** with Jenkins, PyTest, and Mockito across three OS environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket modules** supporting eight workflows and 2M+ synchronized records, improving system reliability across interconnected operational components and user interfaces.
- Automated testing/deployments with **Docker** and Jenkins, reducing deployment time by **50%** while strengthening release confidence and maintainability in real-world runtime environments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **Flask + React** features handling 10K+ daily requests, integrating backend services with user interfaces while maintaining robust code quality under product deadlines.
- Improved **CI/CD quality gates** and concurrent processing controls, reducing turnaround time by **60%** and improving reliability during high-volume production usage.

Skills

Languages: **Python**, **C++**, Java, SQL, JavaScript/TypeScript
Embedded/Connected Systems: firmware-adjacent testing mindset, IoT workflow integration, cloud-connected service validation
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, WebSockets, OOP architecture
Cloud & DevOps: AWS (S3/Glue/RDS), Azure SQL, Docker, Kubernetes, Git, CI/CD pipelines
Quality Engineering: unit testing, integration testing, automation, debugging, performance optimization, Agile/Scrum collaboration

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **connected full-stack workflows** for 50K+ users and 1M+ transactions with real-time updates, resilient integrations, and scalable deployments for production reliability.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **API-driven smart workflows** supporting 100+ daily appointments, adding monitoring-backed validation that improved record retrieval by **2x** and strengthened operational responsiveness.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed a **testable analytics service** across 10+ databases with schema-aware query validation and structured execution checks, improving reliability and maintainability of data operations.